

Homework Section 15.2

1. Evaluate the iterated integral.

a) $\int_1^2 \int_0^1 (1 + 2xy) dx dy$

b) $\int_0^2 \int_0^{\pi/2} x \cos y dy dx$

c) $\int_0^2 \int_0^1 (2x + y)^6 dx dy$

d) $\int_1^6 \int_1^2 \left(\frac{x}{y} + \frac{y}{x} \right) dy dx$

e) $\int_1^2 \int_0^1 (x + y)^{-3} dx dy$

2. Evaluate the double integral.

a) $\iint_R \sin(x + 2y) dA, \quad R = \{(x, y) \mid 0 \leq x \leq \frac{\pi}{2}, 0 \leq y \leq \frac{\pi}{2}\}$

b) $\iint_R \frac{x^2 y}{y^2 + 1} dA, \quad R = \{(x, y) \mid 0 \leq x \leq 1, 0 \leq y \leq 3\}$

c) $\iint_R xye^{x^2 y} dA, \quad R = [0, 2] \times [0, 3]$ **(Hint: integrate w.r.t. x first)**

3. Sketch the solid represented by the integral $\int_0^1 \int_0^1 (2 - x - y) dx dy$.

4. Find the volume of the solid lying between the paraboloid $z = 1 - \frac{x^2}{16} - \frac{y^2}{9}$ and the rectangle $R = [-1, 1] \times [-2, 2]$ that lies in the xy -plane.

5. Find the volume of the solid in the first octant bounded by the cylinder $z = 4 - y^2$ and the plane $x = 1$.